b. 5` GATATG 3` 3` CTACTA 5`
c. 5` GAATTC 3` 3` CTTAAG 5`
d. 5` CACGTA 3` 3` CTCAGT 5`

8. pBR 322, BR stands for
   a. Plasmid Bacterial Recombination
   b. Plasmid Bacterial Replication
   c. Plasmid Boliver and Rodriguez
   d. Plasmid Baltimore and Rodriguez

9. Which of the following one is used as a Biosensors?
   a. Electrophoresis b. Bioreactors
   c. Vectors d. Electroporation

10. Match the following :

Column A	Column B
1 Exonuclease	a. add or remove phosphate
2 Endonuclease	b. binding the DNA fragments
3 Alkaline Phosphatase	c. cut the DNA at terminus
4 Ligase	d. cut the DNA at middle

	1	2	3	4
A)	a	b	c	d
B)	c	d	b	a
C)	a	c	b	d
D)	c	d	a	b

11 In which techniques Ethidium Bromide is used?
   a. Southern Blotting techniques
   b. Western Blotting techniques
   c. Polymerase Chain Reaction
   d. Agrose Gel Electroporosis

12 **Assertion** : Agrobacterium tumifaciens is popular in genetic engineering because this bacteriumis associated with the root nodules of all cereals and pulse crops

   Reason: A gene incorporated in the bacterial chromosomal genome gets atomatically transferred to the cross with which bacterium is associated.
   a) Both assertion and reason are true. But reason is correct explanation of assertion.
   b) Both assertion and reason are true. But reason is not correct explanation of assertion.
   c) Assertion is true, but reason is false.
   d) Assertion is false, but reason is true.
   e) Both assertion and reason are false.

13 Which one of the following is not correct statement.
   a) Ti plasmid causes the bunchy top disease
   b) Multiple cloning site is known as Polylinker
   c) Non viral method transfection of Nucleic acid in cell
   d) Polylactic acid is a kind of biodegradable and bioactive thermoplastic.

14 An analysis of chromosomal DNA using the southern hybridisation technique does not use
   a) Electrophoresis
   b) Blotting
   c) Autoradiography
   d) Polymerase Chain Reaction

15 An antibiotic gene in a vector usually helps in the selection of
   a) Competent cells b) Transformed cells
   c) Recombinant cells d) None of the above

16 Some of the characteristics of Bt cotton are
   a) Long fibre and resistant to aphids
   b) Medium yield, long fibre and resistant to beetle pests
   c) high yield and production of toxic protein crystals which kill dipteran pests.
   d) High yield and resistant to ball worms

17. How do you use the biotechnology in modern practice?

18. What are the materials used to grow microorganism like Spirulina?

19. You are working in a biotechnology lab with a becterium namely E.coli. How will you cut the nucleotide sequence? explain it.

20. What are the enzymes you can used to cut terminal end and internal phospho di ester bond of nucleotide sequence?

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_078-106 CH 04.indd 90 3/14/2020 10:34:01 AM